



INTERNSHIP REPORT

ON

PYTHON-BASED FAQ CHATBOT AND LANGUAGE TRANSLATION TOOL

Submitted To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE

Submitted By

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Batch: 2026-27

1. CANDIDATE'S DECLARATION

I, AYUSHI KUSHWAH, do hereby declare that the internship report titled "Python-Based FAQ Chatbot and Language Translation Tool" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. The work described in this report is based on the projects developed during the internship. I affirm that this report has not been submitted to any other institute for any degree/diploma requirement and I shall be solely responsible for any kind of copyright violation in this regard.

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2. ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their guidance, constructive criticism and friendly advice during this work. I am sincerely grateful to the people who shared their knowledge and helped me understand practical aspects of Python programming, Natural Language Processing and graphical user interface development. I also express my gratitude to my parents for their blessings and continuous support.

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3. INTERNSHIP COMPLETION CERTIFICATE

Attach the internship completion certificate issued by the CODEE ALPHA on this page.



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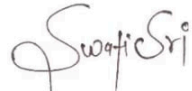
CERTIFICATE OF COMPLETION

This Certificate Is Proudly Presented To

Ayushi Kushwah

This Certificate recognizes the successful completion of the CodeAlpha Virtual Internship Program in Artificial Intelligence, demonstrating dedication and sincerity throughout the program.
We appreciate the valuable efforts and wish continued success in all future endeavors.

10th August 2026 to 10th September 2026



CEO & Co-Founder

10th September 2026

Date of Issue



SCAN TO VERIFY THIS CERTIFICATE
Student ID: CA/DFI/232787

4. PROJECT DESCRIPTION

4.1 Introduction

This internship report presents two Python-based projects: an E-Commerce FAQ Chatbot and a Language Translation Tool. The projects demonstrate practical application of Python programming, Natural Language Processing (NLP), text similarity, graphical user interfaces and external translation services. The chatbot is designed to answer common e-commerce questions by matching a user's query with a prepared FAQ dataset. The translation tool provides a desktop interface where a user can enter text, select source and target languages, translate the text and copy the translated output.

4.2 Organization Profile

The provided internship format identifies the School of Computer Science and Engineering, IILM University, Greater Noida, U.P., and the B.Tech CSE programme. No separate industry/organisation profile was supplied with the project files; therefore, this section is limited to the academic context available in the provided format.

4.3 Problem Statement

Customers frequently ask repetitive questions about orders, returns, payments, shipping and account access. A basic FAQ chatbot can provide quick responses without requiring a human to answer every common query. Similarly, communication across languages can be difficult when users need a quick translation. The second project addresses this need through a simple desktop translation interface.

4.4 Project Objectives

Develop a chatbot capable of matching user questions with predefined FAQ questions.

Apply NLP preprocessing including lowercasing, tokenization, stop-word removal and lemmatization.

Use TF-IDF vectorization and cosine similarity to identify the closest FAQ.

Provide a fallback response when the similarity score is below a defined threshold.

Build a simple translation application with source and target language selection.

Support automatic source-language detection and copying of translated text to the clipboard.

4.5 Scope of the Project

The FAQ chatbot is suitable for a small, predefined set of e-commerce FAQs and can be extended by adding more questions and answers to the dataset. Its current matching approach is based on textual similarity rather than a generative language model. The translation application supports a fixed list of commonly used languages in its interface and uses GoogleTranslator through the deep-translator package. Both projects are suitable as educational prototypes and can be enhanced for larger, production-oriented applications.

4.6 Technologies and Tools Used

Project	Technology/Tool	Purpose
FAQ Chatbot	Python	Core programming
FAQ Chatbot	NLTK	Tokenization, stopwords and lemmatization
FAQ Chatbot	Pandas	FAQ dataset handling
FAQ Chatbot	Scikit-learn	TF-IDF and cosine similarity
FAQ Chatbot	Gradio	Chat interface
Language Translation Tool	Python + Tkinter	Desktop GUI
Language Translation Tool	deep-translator	GoogleTranslator-based translation

4.7 System Architecture

FAQ Chatbot architecture: User Query → NLP Preprocessing → TF-IDF Vectorization → Cosine Similarity with FAQ Questions → Best Match and Threshold Check → FAQ Answer/Fallback Response → Gradio Chat Interface.

Language Translation Tool architecture: User Input → Source/Target Language Selection → Input Validation → GoogleTranslator → Translation Result → Output Display → Copy-to-Clipboard option.

4.8 Methodology

For the FAQ chatbot, a small FAQ dataset is stored as question-answer pairs and converted into a Pandas DataFrame. The questions are preprocessed by converting text to lowercase, tokenizing words, removing punctuation and English stop words, and lemmatizing tokens. The processed questions are converted into TF-IDF vectors. When a user submits a query, the same preprocessing and vectorization steps are applied. Cosine similarity is calculated between the query vector and all FAQ vectors. The highest-scoring FAQ is selected, provided its score reaches the threshold of 0.2; otherwise, the chatbot returns a fallback message.

For the translation tool, a Tkinter window provides source-language and target-language dropdowns, a text input area, a Translate button and an output area. The source can be set to Auto-Detect. The application validates that the input is not empty, initializes GoogleTranslator with the selected source and target languages, and displays the translated result. A copy-to-clipboard button allows the output to be copied when available.

4.9 Expected Outcomes

The expected outcome is a functional demonstration of two practical Python applications. The FAQ chatbot should provide relevant answers for questions similar to the supplied e-commerce FAQs and politely handle out-of-scope queries. The translation tool should provide a simple user-friendly interface for translating entered text between the languages listed in the application. These projects also strengthen understanding of Python modules, data processing, NLP concepts, GUI programming, error handling and application design.

4.10 Certificates of Completion and Other Communication Mail Proof

Attach the internship completion certificate and any official communication/e-mail proof supplied by the internship organisation in this section. These documents were not included in the project files provided for preparation of this report.

5. BIBLIOGRAPHY / REFERENCES

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